

Question 3.3.10:

Construct a right triangle ABC with $AB = 6$ cm, $BC = 8$ cm and $\angle B = 90^\circ$. Draw BD , the perpendicular from B on AC . Draw the circle through B, C and D and construct the tangents from A to this circle.

Solution:**Given**

Let us place the triangle in coordinate geometry using vectors.

Let:

$$\mathbf{b} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{a} = \begin{pmatrix} 6 \\ 0 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 0 \\ 8 \end{pmatrix} \quad (1)$$

Since $\angle B = 90^\circ$, triangle ABC is right-angled at B .

Finding point $\mathbf{d}$: foot of perpendicular from $\mathbf{b}$ to $\mathbf{a} - \mathbf{c}$

$$\text{Let } \mathbf{n} = \mathbf{a} - \mathbf{c} = \begin{pmatrix} 6 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 8 \end{pmatrix} = \begin{pmatrix} 6 \\ -8 \end{pmatrix}$$

Let the general point on line AC be:

$$\mathbf{r} = \mathbf{c} + \lambda(\mathbf{a} - \mathbf{c}) = \mathbf{c} + \lambda\mathbf{n} \quad (2)$$

The foot of the perpendicular from $\mathbf{b}$ to line $\mathbf{r}$ is the point on AC closest to $\mathbf{b}$. Minimize the square of the distance:

$$\|\mathbf{r} - \mathbf{b}\|^2 = (\mathbf{c} + \lambda\mathbf{n} - \mathbf{b})^\top (\mathbf{c} + \lambda\mathbf{n} - \mathbf{b}) \quad (3)$$

Differentiate w.r.t. λ and set derivative to zero:

$$\frac{d}{d\lambda} [\|\mathbf{r} - \mathbf{b}\|^2] = 2\mathbf{n}^\top (\mathbf{c} + \lambda\mathbf{n} - \mathbf{b}) = 0 \quad (4)$$

Solving:

$$\mathbf{n}^\top (\mathbf{c} - \mathbf{b}) + \lambda\mathbf{n}^\top \mathbf{n} = 0 \quad (5)$$

Compute terms:

$$\mathbf{n}^\top (\mathbf{c} - \mathbf{b}) = \begin{pmatrix} 6 & -8 \end{pmatrix} \begin{pmatrix} 0 \\ 8 \end{pmatrix} = -64 \quad (6)$$

$$\mathbf{n}^\top \mathbf{n} = \begin{pmatrix} 6 & -8 \end{pmatrix} \begin{pmatrix} 6 \\ -8 \end{pmatrix} = 36 + 64 = 100 \quad (7)$$

Substitute into 6:

$$-64 + 100\lambda = 0 \Rightarrow \lambda = \frac{64}{100} = \frac{16}{25} \quad (8)$$

Therefore, point $\mathbf{d}$ is:

$$\mathbf{d} = \mathbf{c} + \frac{16}{25}(\mathbf{a} - \mathbf{c}) = \begin{pmatrix} 0 \\ 8 \end{pmatrix} + \frac{16}{25} \begin{pmatrix} 6 \\ -8 \end{pmatrix} = \begin{pmatrix} 8 \cdot \frac{96}{25} \\ 8 - \frac{128}{25} \end{pmatrix} = \begin{pmatrix} \frac{96}{25} \\ \frac{72}{25} \end{pmatrix} \quad (9)$$

*Construct circle through **b, c, d***

Let this be the circumcircle of triangle $\triangle BCD$.

Use determinant form to find center **o** of the circle:

Circumcenter of $\triangle BCD$ lies at the intersection of perpendicular bisectors.

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*Tangents from **a** to circle through **b, c, d***

Let **o** be the center of the circle and r its radius. Then the tangents from **a** satisfy:

$$\|\mathbf{a} - \mathbf{o}\|^2 = r^2 + t^2 \quad (10)$$

Let $\mathbf{t}_1, \mathbf{t}_2$ be points of tangency. The tangent points lie on circle, and vectors $\mathbf{t}_i - \mathbf{o}$ are perpendicular to $\mathbf{a} - \mathbf{t}_i$, so again orthogonality condition applies:

$$(\mathbf{t}_i - \mathbf{o})^\top (\mathbf{a} - \mathbf{t}_i) = 0 \quad (11)$$

This gives quadratic system in coordinates of $\mathbf{t}_i$, which can be solved numerically or geometrically as per need.

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Final Construction Summary

- 1) Plot $\mathbf{b} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\mathbf{a} = \begin{pmatrix} 6 \\ 0 \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} 0 \\ 8 \end{pmatrix}$
- 2) Draw triangle ABC , right-angled at B
- 3) Compute **d** as in 10
- 4) Draw circle through **b, c, d**
- 5) Draw tangents from point **a** to the circle using construction

3D Graph of Triangle ABC with Circle (Q 3.3.10)

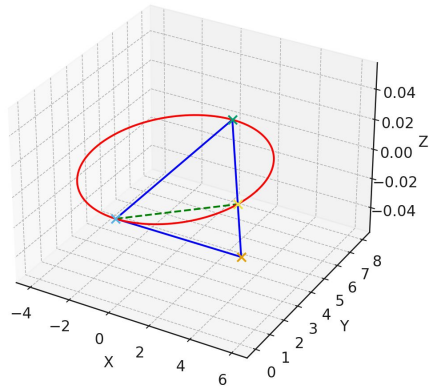


Fig. 1